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Modelling and experimental studies of a water–gas shift catalytic membrane reactor

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HIGHLIGHTS

- A 2D-axisymmetric CMR model was developed using *FLUENT*.
- There was close agreement between the model and the prototype.
- Loading extra catalyst would lead to temperatures beyond recommended ranges.
- Reducing permeation side pressure is a key factor to maximise the CO conversion levels.
- The permeation pressure of the CMR defines the maximum limit of H_2 yield.

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ABSTRACT

A 2-dimensional, axis-symmetric CFD model of a tubular CMR has been developed using a commercial software package *FLUENT* for the purposes of guiding the design and operation of a HTWGS-CMR for the processing of coal-derived syngas. Development of the model has been approached in a stepwise manner through the successive incorporation of sub-models for the CMR processes. For each step, performance of model was checked and validated against measurements using a prototype CMR with same set-up applied in simulation.

The optimum catalyst loading, which yield the maximum CO conversion within targeted operating reactor temperatures (350–450 °C), was found to be 11.6 kg/(CO_{mol}/s) for the inlet syngas temperature of 350 °C with a reactor having a 1″ shell diameter. The CMR model was validated experimentally with a simulated coal-derived syngas (64.5% of CO, 33.0% of H_2 and 2.5% of CO_2 with a 3:1 steam to carbon (S:C) ratio) at a total dry gas flow of 4 L_N/min and a feed pressure of 15 bar_g. These tests were performed using a prototype reactor which incorporated with a tubular (0.1 mm thick, 150 cm^2, 3/8″ OD) Pd/Ag23 wt% membrane. The CMR model was simulated using a wider range of operating parameters (namely permeation rate, inlet temperature, catalyst loading, pressure at permeate side and S:C ratios) to examine its sensitivity to these variables. Outcomes of these parametric analyses have enhanced our understanding of CMR operation in order to optimise its performance.

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1. Introduction

Integrated Gasification Combined Cycle (IGCC) with pre-combustion CO_2 capture offers a means of reducing the CO_2 emissions associated with coal-based power systems. The commercialisation barrier for IGCC, however, is its high capital cost. One means of reducing the capital cost is by process intensification, whereby the water-gas shift (WGS) process and CO_2 separation process are combined. In a conventional system, CO from the syngas is converted into CO_2 via two separate WGS reactors: most of the CO in the syngas is shifted within a high temperature WGS reactor and the small amount of CO left is shifted in a low temperature WGS reactor where operating conditions favour higher equilibrium CO conversion. Both WGS reaction processes and the CO_2/H_2 separation process can be combined in a single catalytic membrane reactor (CMR) using a high-temperature WGS catalyst to achieve CO conversion levels higher than that of the two-step WGS reactor configuration. A previous study demonstrated that WGS conversions of >99% can be achieved at 400 °C compared to the equilibrium value of ~94% for coal-derived syngas with a 3:1 steam to carbon (S:C) ratio [1]. This is explained by the continuous removal of one of the reaction products through the selective membrane which drives the equilibrium of the WGS reaction (Eq. (1)) to the right

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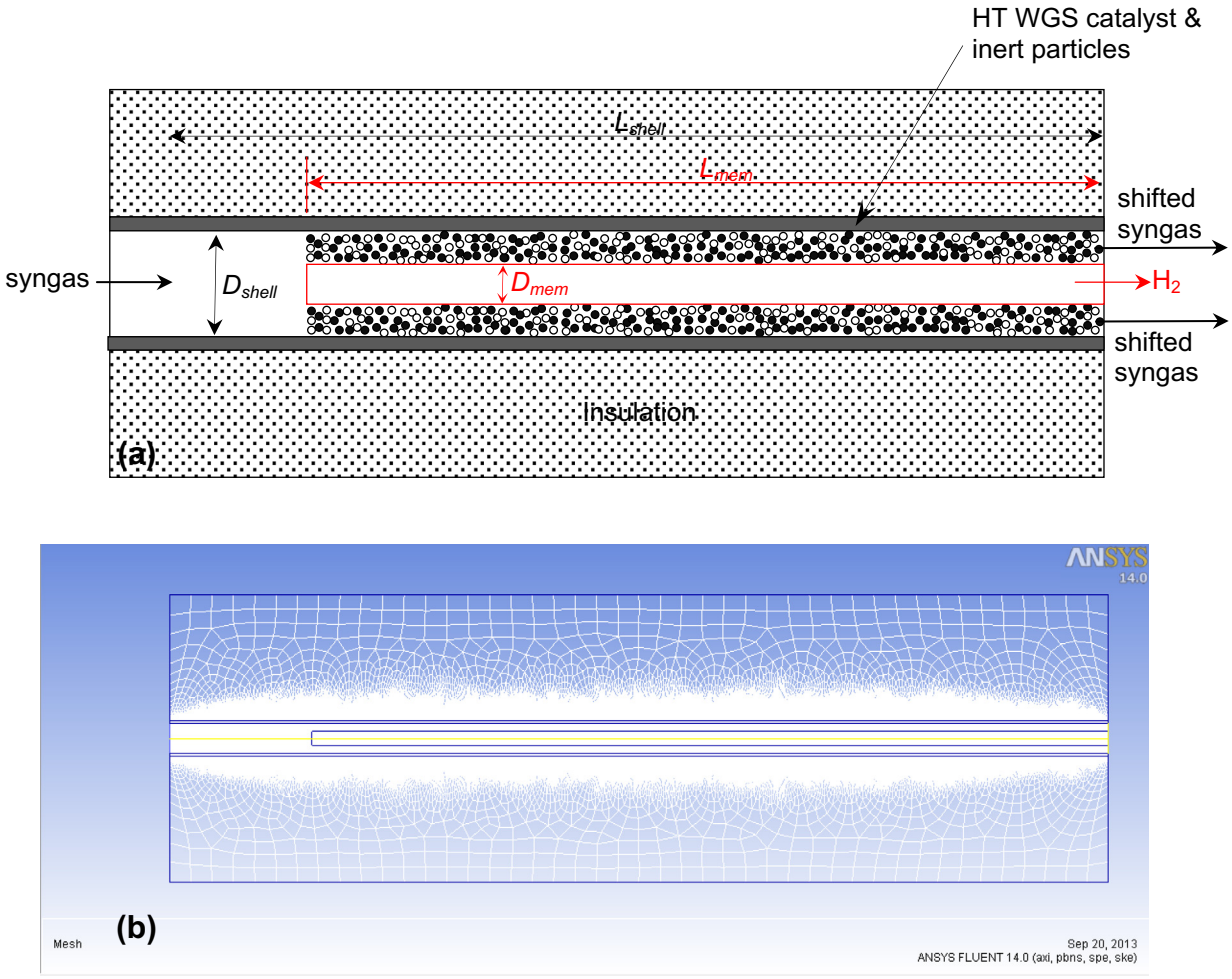


Fig. 1. (a) Schematic diagram of catalytic membrane reactor used in this study. (b) Geometry and mesh of CMR used for 2D-axisymmetric model.

$$CO\ (g) + H_2O\ (g) \leftrightarrow CO_2\ (g) + H_2\ (g)\ \Delta H_{298K} = -41.1\ \text{kJ mol}^{-1} \quad (1)$$

Successful integration of the WGS reaction and the gas separation process into a single unit requires an understanding of heat and mass transfer processes, catalytic WGS reaction rates and H_2 permeation across the membrane under different operating conditions to maximise the advantages of the CMR concept. Mathematical modelling is an effective tool for the design, evaluation and optimisation of any reactors and reaction processes. In the past two decades, a significant number of numerical models have been used to investigate the performance of CMRs [2–29]. Among them, most studies focused on the catalytic water-gas shift reaction inside the CMRs. Some studies modelled for the partial oxidation and catalytic steam-reforming of methane [20–26]. Some developed their models to study the performance of the catalytic membrane reactor for the dehydrogenation of ethylbenzene to styrene [27–29]. Previous models included those with very simplified assumptions that use one dimensional, isothermal and isobaric conditions at steady state excluding the kinetics of the WGS reaction rate [2], an extension to include the pressure drop along the reactor [3,4] and kinetics of the WGS reaction [5]. Other CMR models were relatively more complex taking into account the effects of heat and mass transfer, heat loss, and WGS kinetics but still in one-dimensional in space without considering the radial gradients [6–9]. One-dimensional models can provide fast simulation and quick results with reasonable accuracy compared to two dimensional models [10,11]. Two-dimensional, 2D-axisymmetric or even 3D models, however, are more suitable for H_2 permeation through the membrane in CMRs as this process is truly two-dimensional. Therefore, recent CMR models and membrane module models include the effects of mixing and heat/mass transfer in the radial direction [12–17,30]. Most of the CMR models were developed to study the reactor performance at steady state conditions, whereas only a few of the CMR models were designed to investigate the dynamic behaviour of the CMR [18,19,26].

Over the last decade, CFD modelling with higher space dimensions (i.e., 2D-axisymmetric, 3D) coupled with a detailed understanding of the mechanism and kinetics of the catalytic WGS shift reaction together with mechanisms of H_2 permeation across the membrane has been one of the key tools in understanding of insights into the flow field and temperature profiles within the CMRs that will lead to improving and optimising of its performance. Basile et al. [12] developed a 2D CMR model to investigate the effect of co-current and counter-current flow configurations of CMR. Markatos et al. [13] used *PHOENICS CFD* software to investigate the role of mass and heat dispersion effects on the performance of a large-scale CMR module. Chiappetta et al. [14] investigated the sensitivity of their 2D CMR model in order to define the role of some variables on the performance of a membrane rector for maximising the system efficiency. Chen et al. [15] used a commercial CFD software, *FLUENT*, to develop their WGS reactor model to investigate the performance of both high-temperature and low-temperature WGS reactions. It has to be noted that the H_2 separation process was not included in their model. Byron Smith and Muruganandam [16] also used *FLUENT*

Table 1
Heat transfer sub-models, thermophysical data and transport properties for the solid and gas phases.

Quantity	Sub-models, sources
Specific heat (*Cp*) of individual gas specie (CO, CO_2, H_2O, H_2, N_2)	Temperature dependent (piecewise-polynomial) properties available in ANSYS FLUENT 14.0
Thermal conductivity (*K*) of individual gas specie	Temperature dependent (polynomial) properties available in ANSYS FLUENT 14.0
Viscosity of individual gas specie	Kinetic Theory available in ANSYS FLUENT 14.0
Density of gas mixture	Real-gas-peng-robinson available in ANSYS FLUENT 14.0 [33]
Specific heat of gas mixture	Simple mixing law available in ANSYS FLUENT 14.0 [33]
Thermal conductivity of gas mixture	Ideal gas mixing law available in ANSYS FLUENT 14.0
Mass diffusivity of gas mixture	Kinetic Theory available in ANSYS FLUENT 14.0 [33]
Properties of stainless steel shell	Density = 8030 kg/m^3 [34] *Cp* (J/kg/K) = −1.73E−4 T^2 + 0.406 *T* + 365 [34] *K* (W/m/K) = 1.53E−2 *T* + 9.02 [34]
Properties of HT WGS catalyst	Density = 2100 kg/m^3 *Cp* = 942 J/kg/K *K* = 0.21 (W/m/K)
Properties of insulation material	Density = 100 kg/m^3 *Cp* = 840 J/kg/K *K* (W/m/K) = 2.36605E−07 T^2 −7.17918E−06x + 1.48192E−02

to develop and simulate their 3D CMR model to examine the effects of sweep gas, operating pressures and S:C ratios. Chein et al. [17] examined the effects of several operating parameters of CMR using their 2D model which was developed and simulated using *COMOL* multi physics software. All of these studies were focused on theoretical modelling and no experimental data were used to check the validity of their models except for the work by Chen et al. [15] which used limited experimental data for comparison to the performance of their model which did not include the H_2 separation. Coroneo et al. [30] used limited experimental data to compare with their 3-D model of a Pd–Ag membrane module in which the water-gas shift reaction process was not integrated.

Some of the CMR models in the literature focused on CO conversion and H_2 yield only, and remarkably, did not pay much attention to the temperature profile along the reactor [12,15,16,19,30]. Some simplified 1-D CMR models even assumed the whole process as isothermal, neglecting heat released from the exothermic WGS reaction [2–5]. The temperature profile along the CMR is one of the critical factors when selecting steady-state operating conditions. Alloy membranes can embrittle when temperatures are too low, or suffer degradation from interactions between active and support layers when the temperature is too high. Another factor which limits the upper range of operating temperatures for CMR is the recommended maximum operating temperature of HT WGS catalysts (470–480 °C) [31].

In this study, a 2D-axisymmetric CMR model was developed to achieve better understanding and optimisation of the CMR performance at elevated pressures. One of the main objectives of this reactor-optimization task is to run the CMR within a targeted range of operating temperatures to achieve the highest possible CO conversion and H_2 yield. The CMR model was developed in three steps: (i) a reactor model without integrating both WGS reaction and H_2 separation; (ii) WGS reactor model without H_2 separation and; (iii) the fully developed model that incorporates the WGS and H_2 separation. For each step the validity of model was checked using experimental data.

2. CMR modelling

2.1. Reactor configuration

A schematic diagram of the catalytic membrane reactor used in this study is presented in Fig. 1(a). The system consists of a tubular membrane inside a tubular shell with diameter *D* and a length *L*. Commercial WGS high temperature catalyst, reduced in size to +500–850 μm, was mixed with an inert material of the same size and packed in the annular space between the inner wall of the shell and the outer wall of the membrane. The reactor geometry was constructed for a 2D-axisymmetric model using ANSYS Designmodeler and meshes of the reactor were generated using ANSYS Mesher. As shown in Fig. 1(b) finer meshes were created in the near-wall region to achieve accurate local wall shear predictions: the refined mesh size nearest to the membrane used in the CMR model was as small as 0.14 mm.

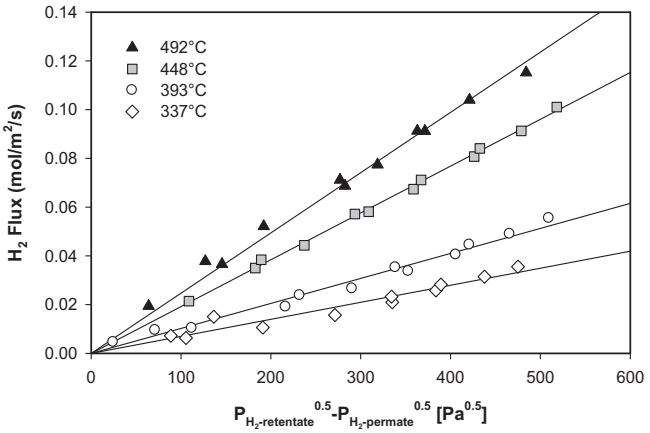


Fig. 2. H_2 permeation fluxes through Pd–Ag membrane against hydrogen partial pressure differences between the retentate and permeate sides at different temperatures.

2.2. Model formulation and properties of gases and solids

The standard transport equations governing 2D-axisymmetric dimensions were derived from the laws of conservation of mass, momentum and energy. Details of those equations can be found in standard heat and mass transfer texts (i.e., Bird et al. [32]) and they are also outlined in the ANSYS FLUENT manual [33]. Gas flow and mixing were modelled by the two equation standard *k*-epsilon model for closure [33]. The thermophysical and transport properties of gas species and gas mixtures are defined using sub-models readily available in *FLUENT*. The physical and chemical properties of solids (i.e., WGS catalyst, insulation material) were defined using the figures provided by manufacturers. The physical–chemical properties as well as the heat and mass transfer parameters used in this study are listed in Table 1.

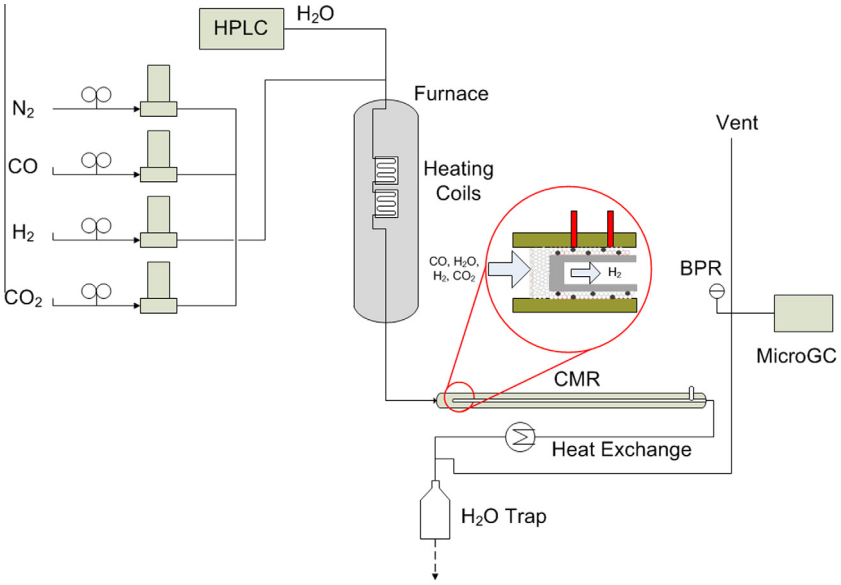


Fig. 3. Schematic diagram of the experimental rig for CMR testing.

2.3. Catalytic WGS reaction sub-model

An accurate estimation of the catalytic WGS reaction rate is critical in any WGS-CMR model as it controls the profile of the reactor temperatures and conversion level of CO. In this study, a power-law reaction rate model proposed by Hla et al. [31] was used to calculate the rate of the WGS reaction over the high temperature catalyst. This reaction model (Eq. (2)) has been applied by others, for example, in a 2D CMR model [17], a transient model of a palladium alloy membrane based catalytic membrane reactor [19], in a 2D dynamic model for packed-bed WGS reactors [35] and modelling of an integrated processor for steam reforming of n-heptane [36]:

$$R = 10^{0.659} \times \exp\left(\frac{-88,000}{R'T}\right) P_{CO}^{0.9} P_{H_2O}^{0.31} P_{CO_2}^{-0.156} P_{H_2}^{-0.05} \left(1 - \frac{1}{K}\frac{P_{CO_2}P_{H_2}}{P_{CO}P_{H_2O}}\right) \quad (2)$$

where R = WGS reaction rate in mol g_{cat}^{-1} s^{-1}, R' = 8.3144 J mol^{-1} K^{-1} and K is an equilibrium constant which was calculated using the equation proposed by Twigg [37] (Eq. (3)):

$$K = \exp[Z\{Z(0.63508 - 0.29353Z) + 4.1778\} + 0.31688] \quad (3)$$

Table 2
Experimental conditions and results using 1″ & 3/4″-shell reactor (loaded with inert material).

N_2 flow (L/min)	H_2O flow	T_{avg} (°C) at TC01	T_{avg} (°C) at TC14*	ΔT (°C)	Heat loss (W)
1″ shell reactor					
4	6.46	349.3	238.1	111.3	34.1
6	9.69	349.2	267.1	82.1	37.9
8	12.92	349.6	283.1	66.5	41.0
10	16.16	350.8	293.5	57.3	44.2
12	19.39	350.8	302.0	48.8	45.3
3/4″ shell reactor					
3.6	5.82	349.5	224.5	124.9	34.4
4	6.46	349.9	233.2	116.7	35.7
6	9.69	349.9	264.0	85.8	39.6
8	12.92	359.1	283.3	65.7	40.5
10	16.16	351.0	298.3	52.7	40.7

* TC14 is located at the end of the CMR.

where $Z = \left(\frac{1000}{T}\right) - 1$, temperature in kelvin.

2.4. H_2 permeation sub-model

An experimental study was conducted to ensure the accuracy of the permeability values of the Pd/Ag23 wt% membrane. A study was conducted using a 0.1 mm thick × 9.95 mm OD × 250 mm long Pd/Ag23 wt% membrane over the temperature range of 300–500 °C, hydrogen partial pressures of 1–8 bar_g in the retentate side and atmospheric pressure on the permeate side of the membrane. The hydrogen flow, measured using an Aalborg mass flow meter that was calibrated using a Brooks 5850S hydrogen mass flow meter and a dry-cal volumetric flow meter, was observed between 18 and 1070 mL min^{-1}.

The hydrogen permeation flux through the Pd/Ag23 wt% membrane can be expressed by means of Eq. (4):

$$J_{H_2} = \frac{P_e}{\delta}\left(P_{H_2,ret}^n - P_{H_2,perm}^n\right) \quad (4)$$

where,

J_{H_2} = the hydrogen diffusion flux through the metal membrane (mol m^{-2} s^{-1})
δ = thickness of membrane (m)
P_{H_2} = hydrogen partial pressure (Pa)
P_e = the hydrogen permeability (mol m m^{-2} s^{-1} $Pa^{-0.5}$)

The temperature dependence of hydrogen permeability (P_e) is defined by Eq. (5):

$$P_e = P_e^\circ \times \exp\left(\frac{-E}{RT}\right) \quad (5)$$

The apparent activation energy (E) and pre-exponential factor (P_e°) of the temperature dependent permeability relationship are essential information to estimate the hydrogen permeation rate across the membrane in the CMR model. In Eq. (4), n is the power dependency of the hydrogen concentration on its partial pressure. n can have a value of between 0.5 and 1 and can be calculated from the correlation between hydrogen pressure and permeation flux [38,39]. If the mass transport of hydrogen is assumed to be controlled by the diffusion of atomic hydrogen, the process can be

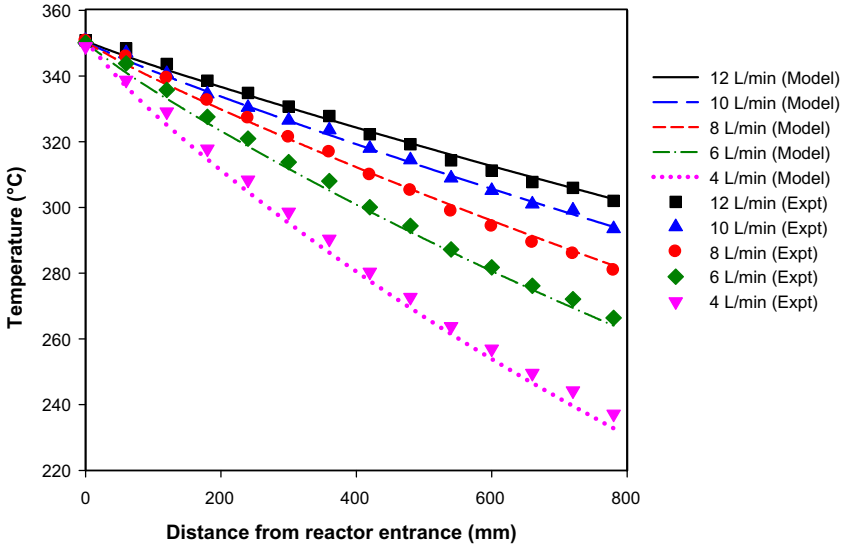


Fig. 4. Experimental data of temperature profiles (inside 1″-shell reactor) compared to predictions using the 2D-axisymmetric model.

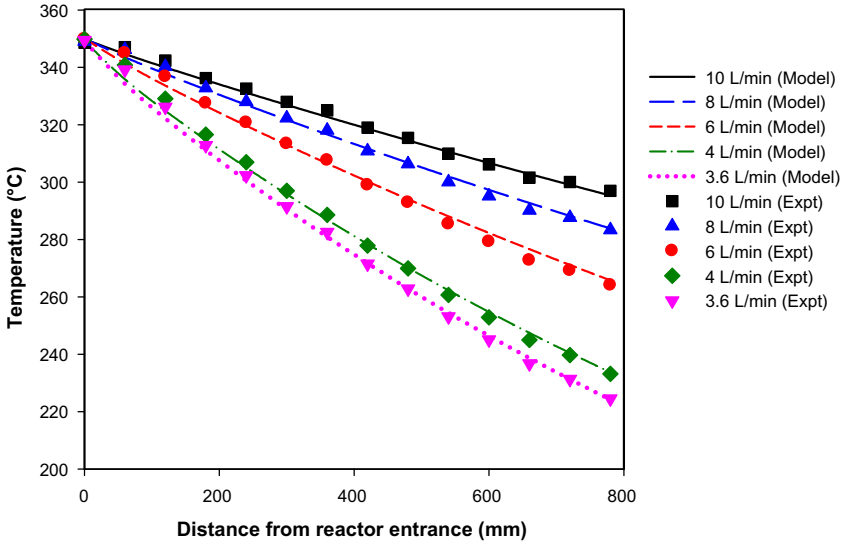


Fig. 5. Experimental data of temperature profiles (inside 3/4″-shell reactor) compared to predictions using the 2D-axisymmetric model.

Table 3
Loading parameters for the CMR conversion studies without membrane.

Length of loading (mm)	Catalyst (g) & wt%	Inert (g)	Syngas flow (L min^{-1})	Gas composition	Pressure, inlet temperature, S:C ratio
200	3.22, 9.5	30.67	4, 6, 8, 10	CO (64.5%)	Pressure = 15 bar(g)
450	7.25, 9.5	69.07	4, 6, 8, 10	H_2 (33.0%)	Temperature = 350 °C
780	12.18, 9.5	116.03	4, 6, 10	CO_2 (2.5%)	Steam:carbon = 3.0

described by the Sieverts-Fick law and the '*n*' factor is taken as 0.5. Fig. 2 shows the correlation between hydrogen permeation rate and the difference in hydrogen pressures between the retentate and permeate sides of membrane assuming an *n* value of 0.5. The data shows linear relationships for all temperatures tested confirming the assumption that *n* is equal to 0.5 and that the diffusion of atomic hydrogen through Pd–Ag membrane is rate limiting and the Sieverts-Fick law is followed. From the experimental data, the pre-exponential factor (P_e° in Eq. (5)) and the apparent activation energy (*E* in Eq. (5)) were determined and the results shown below were used in CMR model:

- P_e° = 7.73E−07 mol m m^{-2} s^{-1} $Pa^{-0.5}$
- *E* = 22,028 J/mol
- for *n* = 0.50

Next, sub-models for determining the water–gas shift reaction rate and the H_2 permeation rate across the Pd–Ag membrane were separately developed in C++ Programming language and exported to *FLUENT* as user-defined functions. All of the governing equations considered in the pressure-based axisymmetric CMR model were then solved simultaneously using ANSYS FLUENT version 14.0.

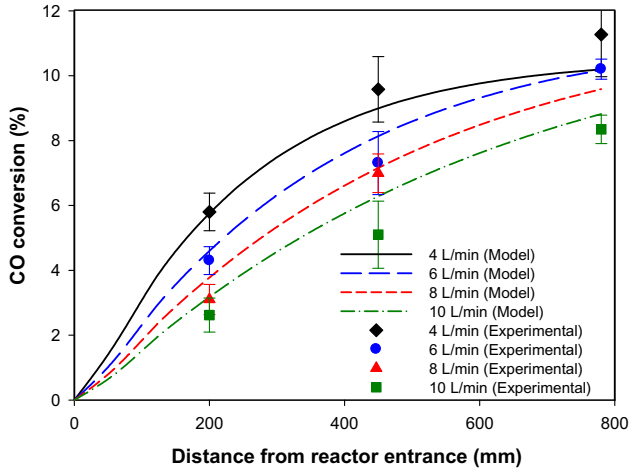


Fig. 6. Comparison between experimental results & model prediction for level of CO conversion under different lengths of catalyst loading and different syngas flows.

3. CMR experimental

3.1. Reactor configuration

A schematic diagram of the experimental rig used in this study is shown in Fig. 3. The system is able to provide temperatures of 1100 °C and pressures up to 16 bar(g). The feed section consists of four individual gas lines and a steam line. CO, H_2, CO_2 and N_2 can be provided to the reactor through mass flow controllers at capacities up to ~10 L/min and a high performance liquid chromatography pump is used to generate a steady flow of de-ionised water for steam production. The gas stream and water injected at the top of a three zone electrical furnace is heated within a 6 m spiral of 1/4″ alloy 600 which increase the residence time and permits the required heat transfer for effective heating of dry gas flows of 12 L/min together with steam (usually set for S:C ratio of 3) at 350 °C. The syngas stream of desired composition and temperature is then delivered to the entrance of the CMR. The tubing from the furnace and the CMR itself has been insulated with 88 mm-thick low thermal conductivity Rockwool Fibertex insulation.

3.2. Measurement system

The syngas transports heat through the CMR and the furnace set-points are optimised to deliver a specified gas temperature at TC01, which is the first thermocouple just prior to the start of the 3/8″ membrane. The temperatures profiles in the CMR are monitored along the length of the CMR using 14 K-type thermocouples installed every 60 mm along the reactor shell wall (in recesses so they do not interfere significantly with the loading of the catalyst or the gas flow). Steam from the outlet gas stream is then removed using a counter current high capacity heat exchanger and the concentrations of product gases (CO, CO_2, H_2 and N_2) are determined by gas chromatography (Varian CP-4900).

3.3. Test programs

The first phase of the experimental program focussed only on heat transfer, without including the WGS and separation processes. The reactor was packed with inert glass beads only, and a non-permeable 3/8″ OD, closed-end stainless steel tube was used in place of the membrane. For the second phase of the program, the reactor was packed with a mixture of WGS catalyst and inert glass beads. The CO conversion level was tested by varying the length and amount of catalyst loading and total syngas flow. The third phase of the program incorporated both the WGS and H_2 separation processes through incorporation of the 150 cm^2 Pd–Ag membrane. Syngas was fed at 4 L/min with a typical syngas composition with a 3:1 S:C ratio. A constant catalyst weight loading was then placed into the annulus along the length of the 540 mm long, 3/8″ OD membrane.

4. Results and discussion

4.1. Reactor performance without WGS reaction and H_2 separation

The 2D-axisymmetric model was first simulated with a feed of N_2 and H_2O only, i.e., excluding the WGS reaction. The purpose of this exercise was to check the validity of heat transfer sub-models used in this study as well as the properties influencing temperature profiles along the reactor, such as the temperature dependent thermal conductivity of the insulation material. The temperature profile is one of the key factors in CMR operation as it defines

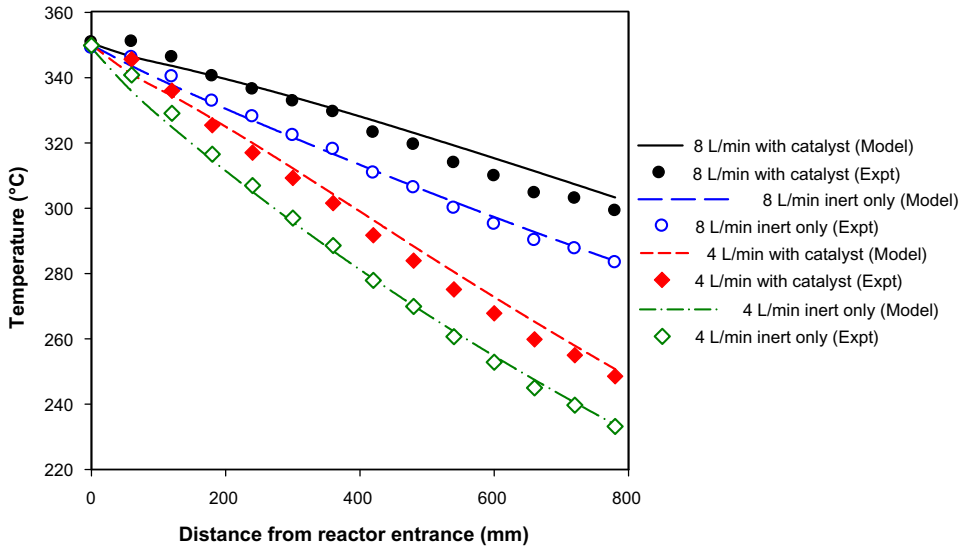


Fig. 7. Comparison between experimental results and model prediction for reactor temperatures under different lengths of catalyst loading and different syngas flow.

Table 4
Different catalyst loadings for the CMR optimisation test.

Simulation no.	Catalyst (wt%)	Inert (wt%)	Catalyst wt (g)	Specific catalyst loading [kg/(CO_{mol}/s)]
Run 1	9.5	90.5	12.57	3.27
Run 2	15.7	84.3	20.30	5.29
Run 3	24.4	75.6	30.45	7.94
Run 4	33.5	66.5	40.40	10.53
Run 5	37.4	62.6	44.46	11.59
Run 6	43.5	56.5	50.65	13.20
Run 7	54.5	45.5	60.60	15.79

the WGS reaction rate and H_2 permeation rate. The same operating conditions were tested using the experimental reactor with shell diameters of 1″ and 3/4″.

Experimental conditions for the first set of tests are listed in Table 2 for both reactors. Figs. 4 and 5 also show the temperature profiles along the 1″-shell reactor and 3/4″-shell reactor compared to profiles predicted using the 2D-axisymmetric model. As expected, significant temperature drops were recorded for smaller flow rates, as the total amount of heat carried by the inlet gases are directly proportional to total mass flow rates. Heat losses along the 780 mm reactor were calculated and presented in Table 2. As expected, greater amounts of heat loss were identified along the CMR for higher inlet flows for both reactors.

Figs. 4 and 5 show that predictions derived from the 2D-axisymmetric model agree very well with experimental data for the wide range of flow rates tested with both reactors. This indicates that the heat and flow sub-models as well as the thermo-physical properties used in this modelling were valid and could be used in the next step of CMR model development which focuses on simulations that integrate the catalytic WGS reaction.

4.2. Reactor performance with WGS reaction excluding H_2 separation

To investigate the CO conversion and temperature profile under catalytic WGS reaction mode, a closed 780 mm stainless steel 3/8″ diameter tube that does not permeate hydrogen was used as a 'blank' in the middle of the tubular CMR. The annular space around the blank membrane was then filled with the catalyst–inert mixture, over three different lengths. Then the reactor was tested under different flow rates of simulated syngas with the other operating parameters (i.e., reactor pressure, inlet temperature, syngas composition and S:C ratio) held constant.

The 2D-axisymmetric model was used to simulate the catalytic WGS reaction where the inlet gas mixture contains a simulated syngas and steam but the H_2 separation process is not included in this step. Input parameters of the WGS simulation runs were the same as the experimental conditions listed in Table 3. The performance of the 2D model with WGS reaction was then compared with experimental data. Figs. 6 and 7 show the profiles of CO conversion and temperature along the reactor respectively, compared with profiles predicted using 2D-axisymmetric model.

In this study, the level of CO conversion is defined as:

$$\text{CO conversion } (\%) = \frac{[CO_{mol(IN)} - CO_{mol(OUT)}]}{CO_{mol(IN)}} \times 100 \quad (6)$$

Fig. 6 shows that CO conversion increases with increasing length of catalyst bed for each of the four flows examined, and the level of CO conversion was less than 10%. As expected the level of CO conversion was observed to decrease with increasing syngas flow. As the amount of catalyst present for all flows was constant for each reactor loading length tested, the specific catalyst loading (which is defined as amount of catalyst per amount of inlet CO mol) decreased when the amount of inlet CO was increased. As a result, a lower level of CO conversion was observed for higher syngas flows under a constant catalyst loading. Fig. 7 shows the increase in temperature profile along the CMR attributed to the WGS reaction and catalyst loading due to heat released from the WGS reaction (which is slightly exothermic). The temperatures inside the reactor, however, did not exceed the inlet temperature by design, in order to explore the accuracy of the model, as the CO conversion levels were low for these series of tests and amount of heat loss was greater than heat produced via WGS reaction.

From Figs. 6 and 7, it can be seen that the predictions using 2D-axisymmetric model agree very well with experimental data for the range of flow rates tested under different lengths of catalyst loading. It has to be noted that the kinetic parameters and mechanism used in this study to estimate the WGS reaction rate were originally derived from low pressure operating conditions [31]. The experimental runs in this study were performed at the conditions of interest (approximately 15 bar(g) of total pressure). The agreement observed between the model prediction and the experimental results at this elevated pressure reveals that a pressure scale-up correlation is not required to apply the power law rate model (derived at low pressure) to the high-pressure operating conditions. This finding agrees with an outcome of a previous study which investigated the effect of total pressure on

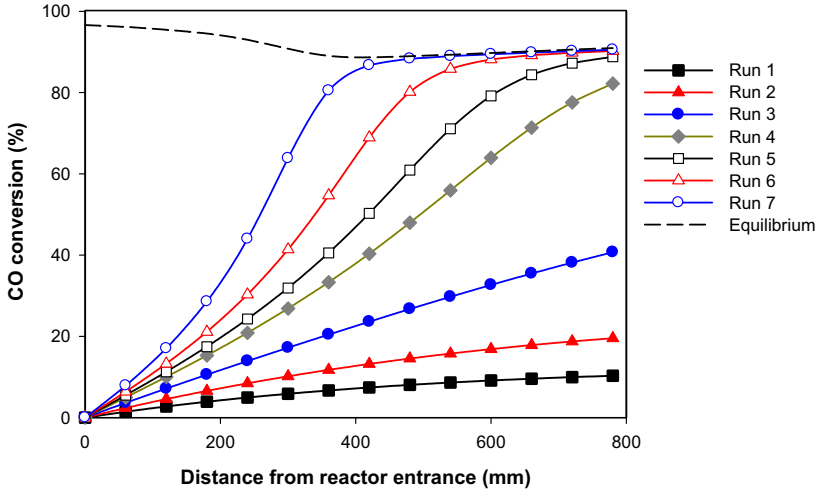


Fig. 8. CO conversion profile for different loadings of WGS catalysts.

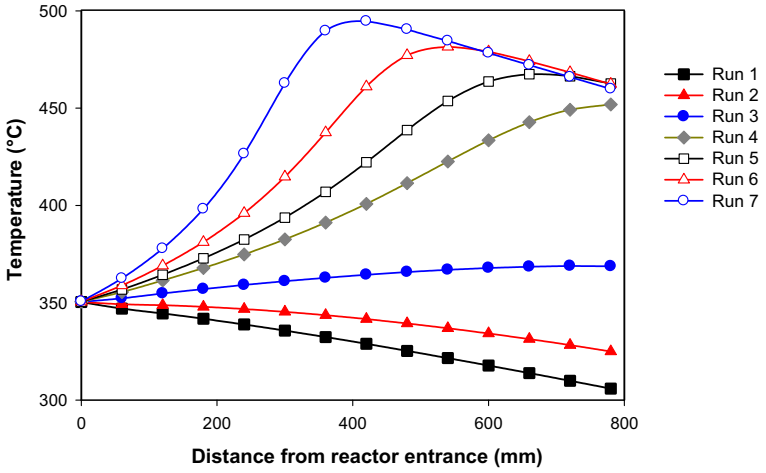


Fig. 9. Temperature profiles for different loadings of WGS catalysts.

Table 5
Experimental results of the CMR test using a tubular Pd–Ag membrane.

	Operating mode	Inlet temperature (°C)	Average CO conversion (%)	H_2 flow at Permeated side (L/min)	H_2 yield at Permeated side (%)
Run 1a	w/o separation	413	92.2	0	0.0
Run 1b	CMR w/o vacuum	413	95.2	1.25	34.8
Run 1c	CMR with vacuum	413	95.7	1.55	43.0
Run 2a	w/o separation	376	87.4	0	0.0
Run 2b	CMR w/o vacuum	376	92.8	0.95	26.9
Run 2c	CMR with vacuum	376	92.4	1.22	34.6
Run 3a	w/o separation	357	25.5	0	0.0
Run 3b	CMR w/o vacuum	357	26.2	0.34	17.5
Run 3c	CMR with vacuum	357	27.2	0.54	27.4

performance of the catalytic WGS reaction using simulated syngases [40]. This demonstrates that the WGS reaction sub-models used in this model are valid and that they can be used to determine the optimum range of catalyst loading for the next step of the modelling process where the CMR model incorporates the H_2 separation process.

4.3. Simulations for optimum catalyst loading

One of the main objectives of the numerical model is to achieve the highest possible CO conversion within a targeted range of operating temperatures. If the operating temperature is higher than 450 °C, it will significantly affect membrane stability and it can also reduce the performance of the HT WGS catalyst [31]. On the other hand, operating the reactor below 300 °C might lead to damage of the membrane caused by hydrogen embrittlement [41]. In this study, therefore, the operating range of CMR temperatures is defined as 350–450 °C with further allowance of ±25 °C. Based on this limitation, the CMR inlet syngas temperature was selected to be 350 °C, as this allows up to 100 °C temperature increases inside the reactor.

The optimum catalyst loading for CMR was investigated by simulating the 2D-axisymmetric model with different catalyst loadings as listed in Table 4. The amount of catalyst loading used for each simulation was normalised by the molar flow rate of inlet CO used in the simulation (listed in the last column of the table).

The simulation results for CO conversion and temperature profile along the reactor for different catalyst loadings are shown in Figs. 8 and 9, respectively. Fig. 8 shows that level of CO conversion approaches equilibrium when the specific catalyst loading was 11.6 [kg/(CO_{mol}/s)] at Run 5. Additional loading of catalyst does not increase the level of CO conversion at the end of the reactor (as the equilibrium CO conversion level is already achieved), but it does increase the peak temperatures inside the reactor as shown in Fig. 9.

At the lower CO conversion levels (~20%), the temperature inside the reactor did not exceed the inlet temperature as the amount of heat loss is greater than heat produced via the WGS reaction. At CO conversion levels of approximately 40% (Run 3), a slightly higher temperature than the inlet temperature was found along the reactor. Further loading of catalyst increases the reactor temperatures as the WGS reaction rate increases exponentially with the increase of temperature and it effectively contributes to a rapid increase in CO conversion until the attainment of equilibrium. Beyond that point, as discussed above, a higher catalyst loading is undesirable as it only produces excessive temperatures inside the reactor and no further increase in CO conversion. Here, it has to be noted that these simulations for defining an optimum range of catalyst loading were done only for a single inlet temperature of syngas with a reactor having a 1″ shell diameter. The effect of inlet temperature and reactor size can significantly change the range of optimum catalyst required for the CMR.

4.4. Reactor performance under full reactor mode

In this configuration, the experimental catalytic membrane reactor was tested incorporating the hydrogen separation process to demonstrate the capability and concept of the CMR tubular configuration using a Pd–Ag membrane. This experimental study was conducted with a 4 L/min syngas flow with the composition mentioned in Table 3 with a 3:1 S:C ratio. A constant catalyst weight (23.96 g of catalyst mixed with 180 g of inert) was then placed into the annular space along the length of the 540 mm-long 3/8″ diameter membrane and 3/4″ diameter shell.

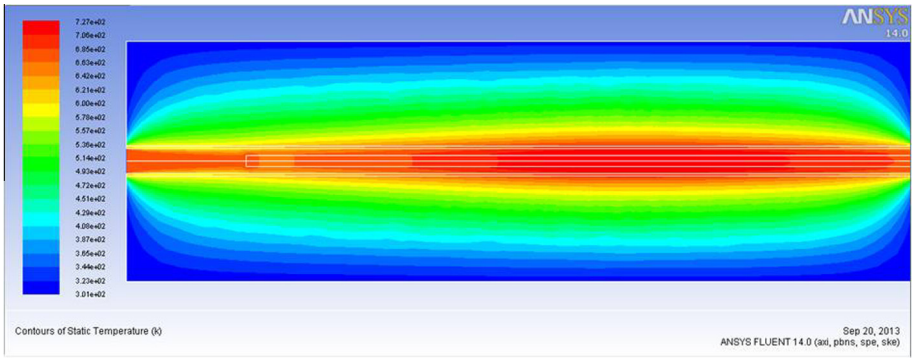


Fig. 10. Temperature contours of Run 2b (inlet temperature (T_{in}) = 376 °C).

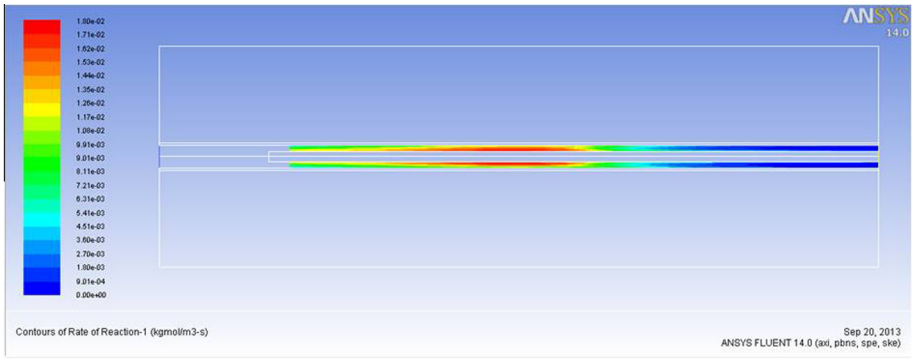


Fig. 11. Contours of water–gas shift reaction rate of Run 2b (inlet temperature (T_{in}) = 376 °C).

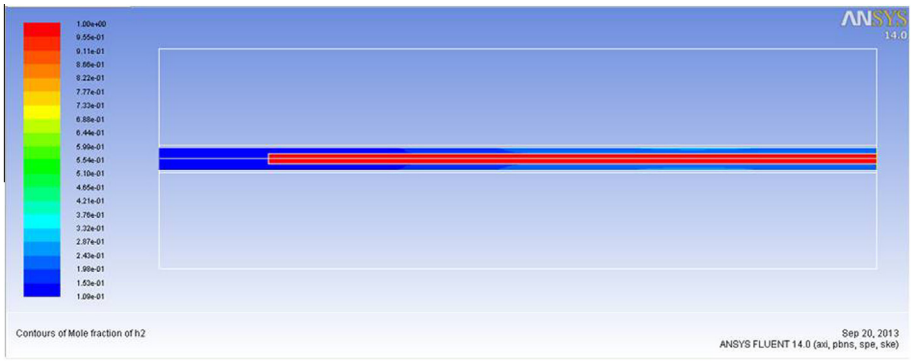


Fig. 12. Contours of H_2 molar fraction of Run 2b (inlet temperature (T_{in}) = 376 °C, with membrane).

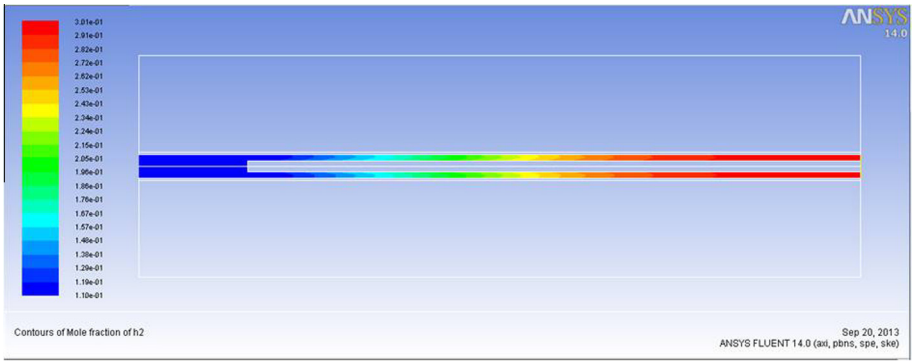


Fig. 13. Contours of H_2 molar fraction of Run 2a (inlet temperature (T_{in}) = 376 °C, without membrane).

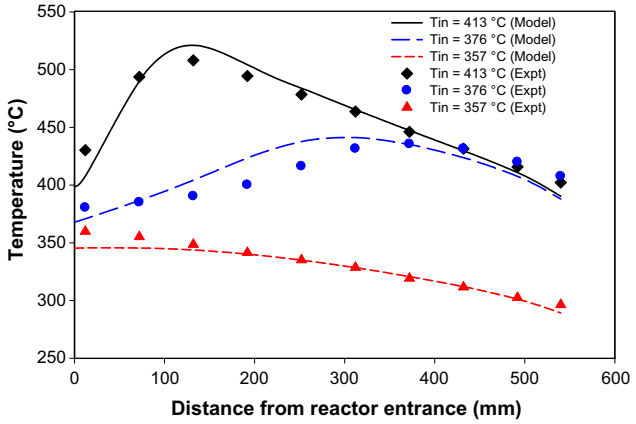


Fig. 14. Effect of inlet temperature (T_{in}) on reactor temperature profiles comparing experimental results (Run 1a, Run 2a & Run 3a) with model predictions.

To test the validity of the model and demonstrate the capability and benefits of the CMR, three inlet temperatures were considered, specifically 413 °C, 376 °C and 357 °C. This range allowed us to consider a wide range of temperature dependent CO conversion and membrane hydrogen permeation rates and hence the model's validity. These tests were conducted with the hydrogen permeation to atmospheric pressure, a vacuum pressure and the path of hydrogen permeation closed (via a valve in the line) to simulate a WGS reaction with non-permeable mode.

The variance in temperature profile along the length of the membrane module, the composition of the converted gas stream, and the rate of the hydrogen flux through the membrane was continuously monitored for each of the temperature set-points and membrane flow conditions. The purity of the permeate was also observed periodically during the 24 h test.

In this study, H_2 yield (%) is defined by Eq. (7):

$$H_2 \text{ yield } (\%) = \frac{H_2(\text{mol})_{\text{out-Permeate}}}{[H_2(\text{mol})_{\text{out-Retentate}} + H_2(\text{mol})_{\text{out-Permeate}}]} \times 100 \quad (7)$$

The experimental conditions and results of these CMR tests are listed in Table 5. Under all three different temperatures tested, integrating the H_2 separation process with WGS reaction promotes CO conversion. At the higher operating temperature, the CO conversion rate increases and it consequently increases hydrogen yield across the membrane. With an inlet temperature of 413 °C, the CO conversion exceeded the equilibrium limit when the separation process was integrated. A further increase in hydrogen yield and CO conversion was observed when a vacuum was applied to the permeate side of the membrane. During the lower inlet temperature tests, both the hydrogen yield and CO conversion were found to be significantly lower, in line with expectations. These experimental results were used to validate the performance of the CMR model.

As discussed previously, the hydrogen permeation equations (Eqs. (4) and (5)) and parameters (which were derived from the permeation experiment described in Section 2.4) were used to model the H_2 separation process in the CMR model. The 2D-axisymmetric model was simulated for the same operating conditions used in the experimental tests listed in Table 5. Simulation results which provide insight into the internal CMR processes (such as contours of temperatures, WGS reaction rates and hydrogen molar fraction) are presented in Figs. 10–13. Fig. 10 shows an increase in reactor temperature due to the heat released by the mildly exothermic WGS reaction as expected and Fig. 11 indicates a peak in the water-gas shift reaction rate occurs 150–250 mm from the entrance of the CMR.

Figs. 12 and 13 show the contours of molar fraction of H_2 for Run 2b and Run 2a which represents the CMR operating with and without H_2 separation, respectively. These two figures clearly show the differences in hydrogen molar fraction along the retentate and permeate sides of the CMR when the CMR operates with or without a membrane.

Key model results taking the mass-weighted average values across the retentate and permeate side of the reactor are shown in Figs. 14–17 and compared with experimental results. The effect of reactor inlet temperature on the reactor temperature profile, CO conversion profile and hydrogen yield are presented Figs. 14–16, respectively. Fig. 14 shows that the reactor temperature was lower than the inlet temperature when the inlet temperature was 357 °C. However, when the inlet temperature was increased to 376 °C, a significant increase in temperature profile was observed along the reactor, due to the exponential nature of the WGS reaction which promotes WGS reaction rate rapidly when heat produced via the WGS reaction is greater than the amount of heat loss. A further increase in the inlet temperature increases the peak temperatures inside the reactor, but the increase in CO conversion at the exit of the CMR is small as the level of CO conversion is already close to equilibrium as shown in Fig. 15. At the higher and more representative operating temperature, the CO conversion rate increased and it consequently increased the hydrogen pressure on the retentate side. As described in Eqs. (4) and (5), the elevated

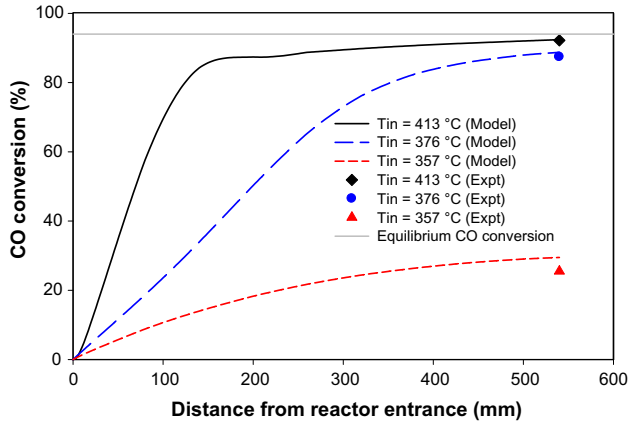


Fig. 15. Effect of inlet temperature (T_{in}) on CO conversion comparing experimental results (Run 1a, Run 2a & Run 3a) with model predictions.

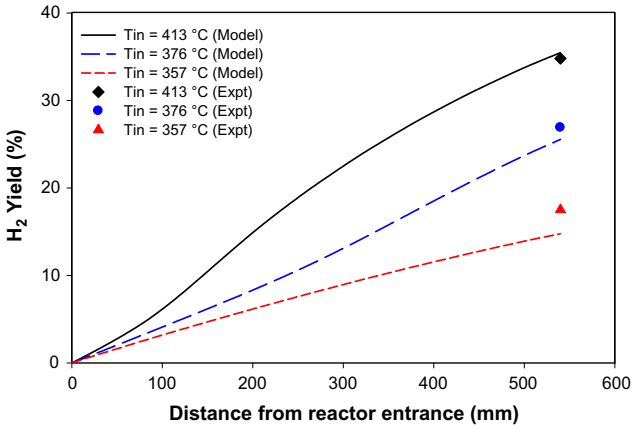


Fig. 16. Effect of inlet temperature (T_{in}) on H_2 yield comparing experimental results (Run 1b, Run 2b & Run 3b) with model predictions.

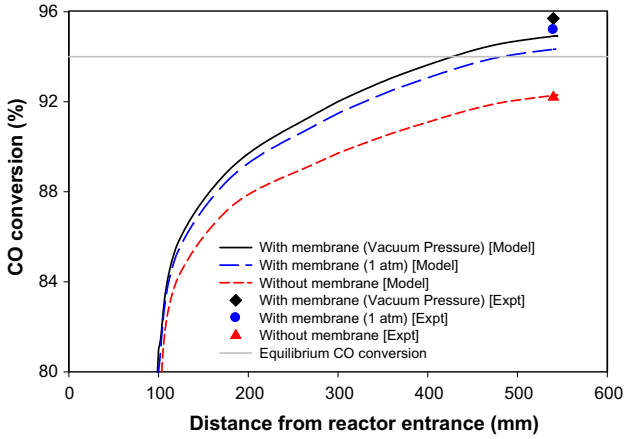


Fig. 17. Effect of H_2 separation on CO conversion comparing experimental results (Run 1c, Run 1b & Run 1a) with model predictions at T_{in} = 413 °C.

operating temperature and higher hydrogen partial pressure on the retentate side of the membrane increase the permeation rate. As a result, an increase in hydrogen yield was observed at higher inlet temperatures as shown in Fig. 16.

The effect of hydrogen separation on CO conversion and hydrogen yield are presented in Fig. 17. When the reactor was operating without a membrane, the CO conversion profile was solely controlled by the WGS reaction rate and its maximum possible level was limited by thermodynamic equilibrium. When the CMR was operated with a membrane, the continuous removal of hydrogen from the retentate side of the reactor promotes the forward reaction, and a CO conversion level higher than the equilibrium value (without integrating separation process) can be achieved, as shown in Fig. 17. This shows that further increases in hydrogen yield and CO conversion level can be achieved by reducing the hydrogen partial pressure on the permeate side of the membrane.

4.5. Parametric analysis on reactor performance

The main purpose of the numerical model is to maximise the reactor performance by finding the optimum range of operating parameters. It is therefore important to investigate the effect of variation in these parameters upon model performance. A parametric sensitivity analysis was therefore carried out to explore the effects of variations in operating and model parameters on the level of CO conversion and H_2 yield.

To check the sensitivity of the model, the following operating parameters were investigated, namely (i) amount of catalyst loading, (ii) inlet temperature of syngas and steam mixture, iii) pressure of permeation side and (iv) S:C ratio – all as a function of hydrogen permeation rate (through the use of a 'permeation multiplication factor'). The parametric studies were conducted by changing the value of one specific parameter while all the other model parameters were kept constant at the base level given in Table 6. In this study, a relatively thick (0.1 mm) membrane has been used to validate the CMR model, as discussed in Section 2.4. The resultant permeation will be lower than more advanced thin membranes, but thinner membranes are far more variable in their permeation properties due to variations in fabrication, thickness and the influence of support structures. To check the effect of the greater H_2 permeation rates on performance of the CMR, the pre-exponential factor ($P_e°$ for Eq. (5)) was increased in this parametric analysis by up to a factor of ten over the baseline value.

Table 6
Baseline operating parameter values used for sensitivity analysis.

Inlet temperature of syngas	375 °C
Amount of Catalyst loading	12 kg s/CO_{mol}
Pressure of permeation side	1 atm
Steam to carbon (S:C) ratio	3
Pre-exponential factor ($P_e°$)	7.73E−07 mol m m^{-2} s^{-1} $Pa^{-0.5}$

4.5.1. Effect of catalyst of loading

The effect of catalyst loading on CO conversion, with a range of hydrogen permeation rates, is shown in Fig. 18(a). As expected the higher catalyst loading increased the CO conversion levels for all H_2 permeation rates. Beyond a certain level of catalyst loading, however, no significant increase in CO conversion was achieved. For example, a signification increase in CO conversion was predicted under all H_2 separation conditions when the catalyst loading was increased from 10 kg s/CO_{mol} to 12 kg s/CO_{mol}. Any further addition of WGS catalyst beyond 16 kg s/CO_{mol} did not yield further significant improvement in CO conversion level. One of the reasons for this is that beyond a certain level of catalyst loading, the level of CO conversion approaches the equilibrium CO conversion level which limits further increase when the H_2 separation process is excluded. When the permeation multiplication factor is equal to zero on the X axis in Figs. 18–21, it is equivalent to the operating condition of no H_2 separation process. It can be seen from Fig. 18(a) that H_2 separation promotes CO conversion beyond the equilibrium limit. For the inlet temperature of the base case, CO

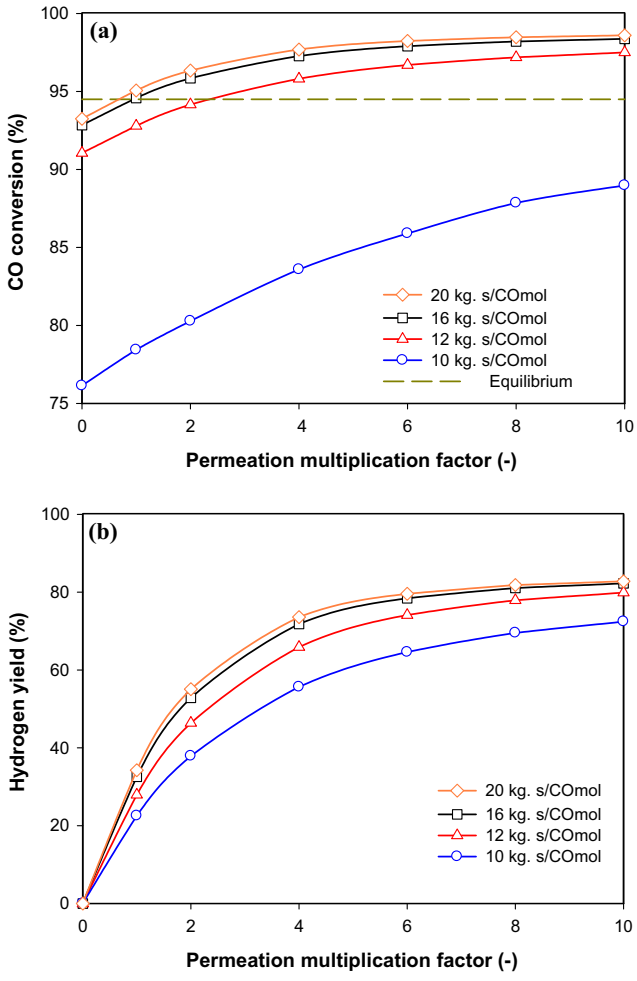


Fig. 18. Effects of catalyst loading and H_2 separation rate on (a) CO conversion & (b) H_2 yield (%).

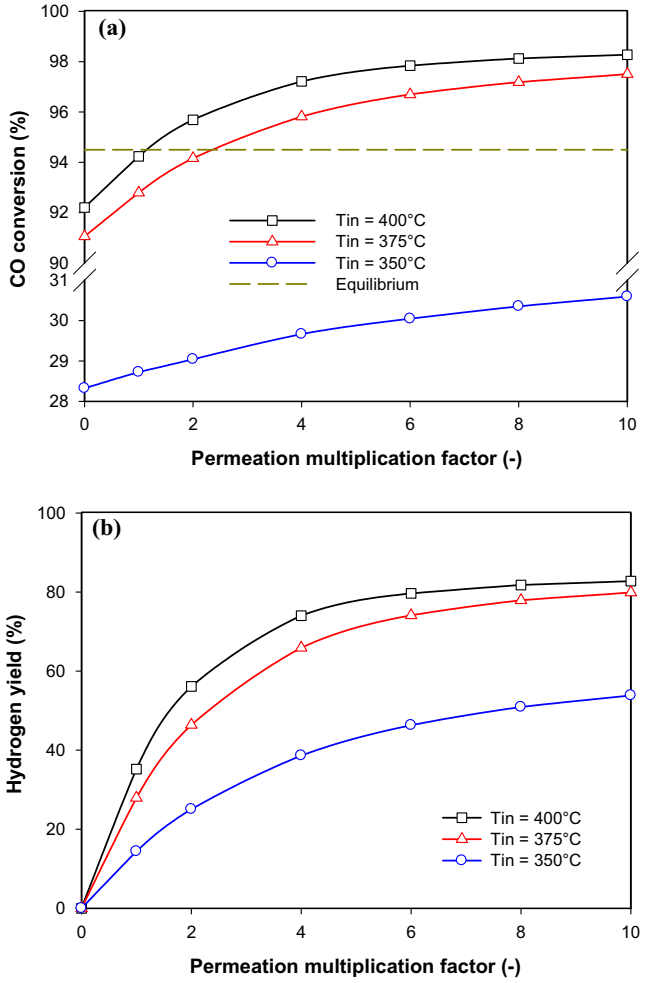


Fig. 19. Effects of inlet temperature (T_{in}) of syngas and H_2 separation rate on (a) CO conversion & (b) H_2 yield (%).

conversion in excess of 98% can be attained when the permeation multiplication factor is greater than 6 (i.e., $P_e^{\circ} \geqslant 4.46E{-}06$ mol $m/m^2/s/Pa^{0.5}$). As expected, H_2 yield depends upon the extent of hydrogen separation. In general, the higher rate of H_2 permeation via the membrane increases the H_2 yield of the CMR. The increase of H_2 yield was found to be more significant when the permeation multiplication factor was increased from 1 to 6 and further increases in the permeation multiplication factor do not have much impact on hydrogen yield especially for those cases with higher catalyst loading.

4.5.2. Effect of inlet temperature

The inlet temperature is one of the critical operating parameters which determines the CO conversion and H_2 yield of the CMR. As the effect of inlet temperature upon the CMR temperature profile has already been discussed in Section 4.4, it will not be repeated here. Fig. 19(a) and (b) show the effects of inlet temperature on CO conversion and H_2 yield respectively under different H_2 separation rates. As the WGS reaction is an exothermic reaction which increases exponentially with temperature, a small increase in inlet temperature can significantly promote the level of CO conversion. As shown in Fig. 19(a), when the inlet temperature was 350 °C, CO conversion was found to be as low as 28%. For the case of 375 °C of inlet temperature, CO conversion level increased to 91%. This exponential increase was already confirmed by the experimental results as discussed in Section 4.4. The effect of changing inlet temperature on CO conversion is similar to the effect of changing the catalyst loading, as both parameters have the same effect on CMR operating temperatures. As discussed in Section 4.5.1, when the CO conversion level approaches equilibrium (for the operating mode without separation) and CO conversion level approaches 100% (for the operating mode with separation), a further increase of inlet temperature does not lead to a higher CO conversion. A CO conversion close to 98% was achieved by increasing the permeation multiplication factor by 6 for the case of inlet temperature of 400 °C. Compared to the case with a high amount of catalyst loading (20 kg s/CO_{mol}) (see Fig. 18(b)), a similar high level of hydrogen yield (approximately 83%) was achieved by increasing the inlet temperature to 400 °C and the permeation multiplication factor to 10 (see Fig. 19(b)).

4.5.3. Effect of pressure of permeation side

The effect of permeation-side pressure on CO conversion and H_2 yield for different H_2 permeation rates was plotted in Fig. 20(a) and (b), respectively. Reducing the permeation pressure was found to be an effective way to achieve a higher hydrogen yield. According to Eq. (4), the hydrogen partial pressure on the permeate-side is a limiting factor for hydrogen yield. In the base case of this sensitivity analysis, the permeate-side pressure was set at 1 atm. At this pressure, a hydrogen yield of approximately 83% was achieved by increasing catalyst loading and by increasing inlet temperatures at the maximum permeation multiplication factor (=10), as shown in Figs. 18(b) and 19(b). A reduction in the

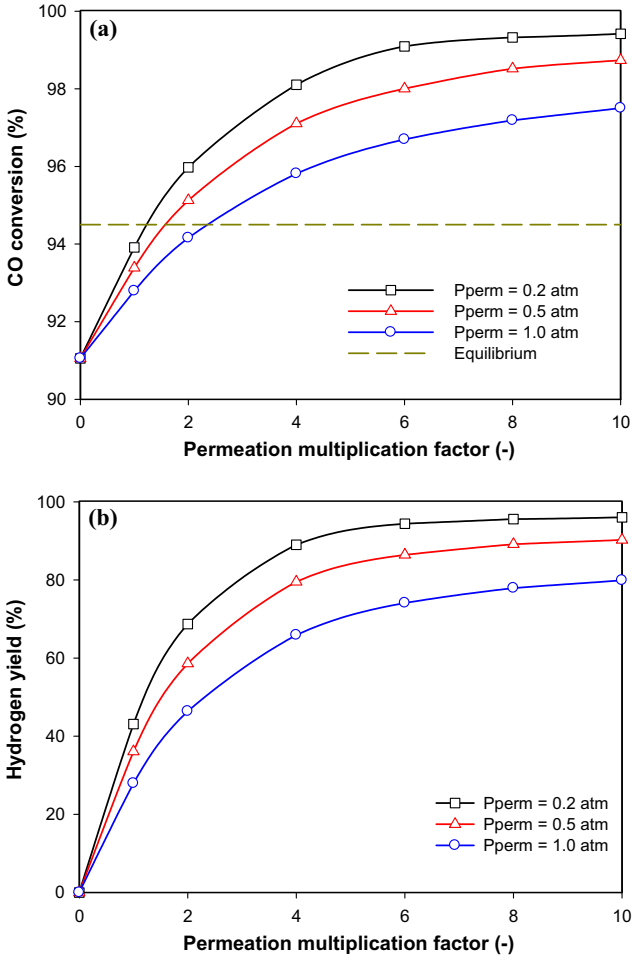


Fig. 20. Effects of pressure of permeation side and H_2 separation rate on (a) CO conversion & (b) H_2 yield (%).

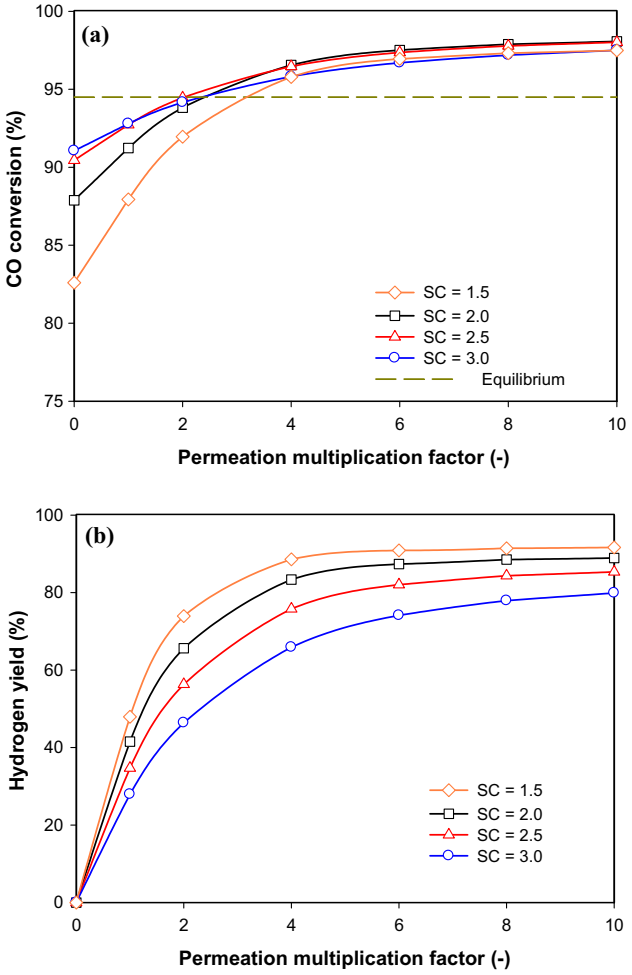


Fig. 21. Effects of S:C ratio and H_2 separation rate on (a) CO conversion & (b) H_2 yield (%).

permeation side gas pressure to 0.5 bar(g) and 0.2 bar(g) resulted in significant increase in H_2 yield to 90% and 96% (at $P_e^\circ = 10$) respectively, as shown in Fig. 20(b). The rate of H_2 removal by the membrane is the only factor that determines the extent of CO conversion beyond the equilibrium limit. A very high H_2 yield and a CO conversion over 99% were achieved by reducing the permeation side pressure to 0.2 bar(g) and increasing the permeation multiplication factor beyond 6. This analysis shows that permeation side pressure is the one of the key parameters for achieving high levels of H_2 yield and CO conversion for a given H_2 membrane. This can be achieved practically through the use of a sweep gas like nitrogen when a turbine is going to be used to generate electricity or a vacuum where a pure hydrogen stream is still required.

4.5.4. Effect of steam to carbon (S:C) ratio

A lower S:C ratio in the WGS process is favourable as it means the energy penalty associated with steam generation is reduced. A large excess of steam (S:C ratios of 3–5), however, is usually recommended to avoid carbon formation caused by the disproportionation of CO via the Boudouard reaction [42]. In this study, therefore, both experimental tests and model simulations reported to this point were undertaken with a constant S:C ratio of 3:1 in order to avoid any possible carbon formation. Sensitivity of S:C ratio was analysed by varying it from 1.5 to 3. For syngas with a typical composition, a higher equilibrium CO conversion is achieved by increasing S:C ratio, as expected. This explains why the highest CO conversion is obtained for the highest S:C ratio simulated in this analysis when the H_2 separation process is excluded (i.e., permeation multiplication factor = 0) in Fig. 21(a). However, when the H_2 separation process was integrated to the system, the CO conversion levels increased more significantly for those simulations using the lower S:C ratios as shown in Fig. 21(a), due to their higher H_2 permeation rates. Eq. (4) states that a higher H_2 permeation rate can be achieved by either increasing H_2 partial pressure in the retentate side or decreasing H_2 partial pressure in permeate side (for a given pre-exponential factor and the apparent activation energy). At the lower S:C ratio for a given total system pressure (i.e., 15 bar(g) in this study), the partial pressure of steam is reduced, consequently the partial pressures of syngas gases species including H_2 increase. As a result, H_2 permeation rates are promoted by lowering S:C ratios and, higher hydrogen yields were achieved for smaller S:C ratios as shown in Fig. 21(b). It has to be noted here that carbon formation is a serious issue usually found in operating condition at lower S:C ratio and, this impact has to be considered alongside the goals of achieving high H_2 yield and CO conversion.

5. Conclusions

A CFD model of a high-temperature water-gas shift catalytic membrane reactor (HTWGS-CMR) has been developed and simulated using a commercial software package *FLUENT*. In order to validate this model, a prototype HTWGS-CMR has been constructed and tested in three modes of operation (no WGS and separation, WGS only and WGS + separation). There was close agreement between the model and the experimental results, confirming that the heat transfer and fluid flow sub-models, mechanism and kinetic parameters used to estimate catalytic WGS reaction rates and H_2 permeation rates used in this study were valid.

The results demonstrate that the thermodynamically limited equilibrium CO conversion level can be overcome by incorporating a hydrogen-selective membrane into a catalytic HT-WGS reactor. In general, higher CO conversion levels and H_2 yields can be achieved by increasing inlet temperatures, loading more catalyst, increasing the H_2 permeation rate, reducing pressure in permeate side of the reactor and reducing the S:C ratio. However, increasing the inlet temperatures and loading more catalyst beyond a particular level would lead to temperatures that are beyond the recommended operating temperatures for long term operation of membrane and HTWGS catalysts. Promoting performance of membrane increases H_2 yields which also raise the CO conversion levels. However, the permeation pressure of the CMR defines the maximum possible limit of H_2 yield. Reducing the pressure at the permeation side of the CMR is one of the effective ways to increase the CO conversion levels close to 100%. Reducing the S:C ratio can also improve the performance of the CMR but this is only valid down to a certain level as low S:C ratios can lead to carbon formation.

This model now provides a basis for evaluating reactors under varying operating conditions and H_2 separation rates, thereby reducing the experimental burden associated with CMR development.

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